

.1. p -adic Numbers and p -adic Integers

Fix a prime number p .

Definition .1.1

For $x \in \mathbb{Z} \setminus \{0\}$, define the p -adic valuation

$$v_p(x) = \max\{k \in \mathbb{Z}_{\geq 0} : p^k \mid x\}.$$

By convention, set

$$v_p(0) = \infty.$$

Remark .1.1

If $x, y \in \mathbb{Z}$, then

$$v_p(xy) = v_p(x) + v_p(y).$$

Hence for $x, y \in \mathbb{Q}^\times$, writing $x = a/b$ with $a, b \in \mathbb{Z}$ and $b \neq 0$, we extend v_p by

$$v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b).$$

For example,

$$v_2\left(\frac{3}{32}\right) = -5.$$

Definition .1.2

For $x \in \mathbb{Q}^\times$, define the p -adic absolute value by

$$|x|_p = p^{-v_p(x)},$$

and set $|0|_p = 0$.

Example .1.1

$$\left|\frac{3}{32}\right|_2 = 2^5 = 32, \quad \left|\frac{1}{2^{100}}\right|_2 = 2^{100}, \quad |2^{100}|_2 = 2^{-100} = \frac{1}{2^{100}}.$$

Proposition .1.1

The function

$$d(x, y) = |x - y|_p$$

defines a metric on $\mathbb{Q}$.

Proof. We have the following facts:

- $|x|_p = 0$ if and only if $x = 0$;
- $|xy|_p = |x|_p |y|_p$;

- for all $x, y \in \mathbb{Q}$,

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

The last inequality is the non-Archimedean triangle inequality, and it implies the ordinary triangle inequality as well.

The proof of these facts are easy to check, for the last one, write $x = p^m \frac{r}{s}$, $y = p^n \frac{u}{v}$, where $r, s, t, u \in \mathbb{Z}$ but no divisible by p and $m, n \in \mathbb{Z}$. Then let $k = \min(m, n)$.

$$v_p(x + y) = k + v_p(x' + y') \geq k$$

where $x' = \frac{x}{p^k}$ and $y' = \frac{y}{p^k}$. Therefore,

$$|x + y|_p = p^{-v_p(x+y)} \leq p^{-k} = \max\{|x|_p, |y|_p\}$$

$\mathbb{Z}$

Example .1.2

If $x = 32$ and $y = 48$, then

$$|x|_2 = \frac{1}{32}, \quad |y|_2 = \frac{1}{16},$$

so

$$|x + y|_2 = |80|_2 = \frac{1}{16} \leq \max\left\{\frac{1}{32}, \frac{1}{16}\right\} = \frac{1}{16}.$$

Definition .1.3

Let (X, d) be a metric space.

- A sequence (x_n) in X is a *Cauchy sequence* if

$$\lim_{m, n \rightarrow \infty} d(x_m, x_n) = 0.$$

- Two Cauchy sequences (x_n) and (y_n) are *equivalent* if

$$\lim_{n \rightarrow \infty} d(x_n, y_n) = 0.$$

Definition .1.4

The *p-adic numbers* $\mathbb{Q}_p$ are defined to be the completion of $\mathbb{Q}$ with respect to the metric induced by $|\cdot|_p$. This means we add all the limits of the Cauchy sequences to $\mathbb{Q}$.

Proposition .1.2

We have the following properties of $\mathbb{Q}_p$:

- $\mathbb{Q}_p$ is a field.
- $\mathbb{Q}_p$ is complete.
- $\mathbb{Q}$ is dense in $\mathbb{Q}_p$.

Proof.

Write the class of Cauchy sequence (x_n) as $[(x_n)]$. Define addition and multiplication as

$$\begin{aligned} [(x_n)] + [(y_n)] &:= [(x_n + y_n)] \\ [(x_n)] \cdot [(y_n)] &:= [(x_n \cdot y_n)]. \end{aligned}$$

Well-defined:

Suppose we have

$$(x_n) \sim (x'_n) \quad (y_n) \sim (y'_n).$$

then

$$|(x_n + y_n) - (x'_n + y'_n)|_p \leq \max(|x_n - x'_n|_p, |y_n - y'_n|_p)$$

which goes to 0. For multiplication,

$$\begin{aligned} |x_n y_n - x'_n y'_n|_p &= |x_n(y_n - y'_n) + y'_n(x_n - x'_n)|_p \\ &\leq \max(|x_n|_p |y_n - y'_n|_p, |y'_n|_p |x_n - x'_n|_p) \end{aligned}$$

Since Cauchy sequence are bounded, both term goes to 0.

The field axioms are easy to check expect for inverse. Choose $[(x_n)] \neq 0$. We check that $[(x_n^{-1})] \in \mathbb{Q}_p$. It suffices to show that (x_n^{-1}) is Cauchy.

$$\begin{aligned} |x_n^{-1} - x_m^{-1}|_p &= \left| \frac{x_m - x_n}{x_n x_m} \right|_p \\ &= \frac{|x_m - x_n|_p}{|x_n|_p |x_m|_p} \end{aligned}$$

Since the denominator is bounded and does not take 0 for sufficiently large m, n . Thus it goes to 0.

So $\mathbb{Q}_p$ is a field.

- The second statement can be proved with some diagonal argument.
- Let d^* be the metric in $\mathbb{Q}_p$. For any $[(y_n)]$ in $\mathbb{Q}_p$, choose N such that $|y_n - y_m| < \frac{\epsilon}{2}$ for $m, n > N$. Let $x_n = y_N$ for all n . Then

$$\lim_{n \rightarrow \infty} |y_n - x_n|_p < \epsilon$$

So $\mathbb{Q}$ is dense in $\mathbb{Q}_p$.

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Definition .1.5

The p -adic integers are

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}.$$

Equivalently, $\mathbb{Z}_p$ is the completion of $\mathbb{Z}$ with respect to $|\cdot|_p$.

Remark .1.2

If $x \in \mathbb{Z}$ and $p \nmid x$, then $|x|_p = 1$, so

$$\left| \frac{1}{x} \right|_p = 1.$$

For example,

$$\frac{1}{3} \in \mathbb{Z}_2.$$

Proposition .1.3

Every element of $\mathbb{Q}_p$ can be written in the form

$$x = p^{-n}u$$

for some $n \in \mathbb{Z}_{\geq 0}$ and some $u \in \mathbb{Z}_p$.

HW:

Show that

$$p\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p < 1\},$$

that $p\mathbb{Z}_p$ is the unique maximal ideal of $\mathbb{Z}_p$, and that

$$\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}.$$

Definition .1.6

A *coherent p -adic sequence* is a sequence

$$a_1, a_2, \dots$$

such that

$$0 \leq a_n \leq p^n - 1$$

for all n , and

$$a_n \equiv a_{n-1} \pmod{p^{n-1}}$$

for all $n \geq 2$.

Example .1.3

If $p = 3$, then

$$1, 4, 22, 49, \dots$$

is a coherent 3-adic sequence.

Theorem .1.4

Every $x \in \mathbb{Z}_p$ can be written uniquely as

$$x = b_0 + b_1p + b_2p^2 + \cdots$$

with

$$b_i \in \{0, 1, \dots, p-1\}.$$

Remark .1.3

The series converges in $\mathbb{Z}_p$ because its partial sums

$$a_n = b_0 + b_1p + \cdots + b_np^n$$

form a Cauchy sequence in the p -adic metric.

Remark .1.4

Distinct choices of digits b_i give distinct elements of $\mathbb{Z}$. Moreover, for each $x \in \mathbb{Z}$, one may choose digits recursively by requiring

$$\begin{aligned} b_0 &\equiv x \pmod{p}, \\ b_0 + b_1p &\equiv x \pmod{p^2}, \\ &\vdots \end{aligned}$$

Hence there is a bijection between $\mathbb{Z}$ and coherent p -adic sequences.

Remark .1.5

In the notation above, $x \in \mathbb{Z}_p$ is a unit if and only if $b_0 \neq 0$.

Proposition .1.5

Every $x \in \mathbb{Q}_p \setminus \mathbb{Z}_p$ can be written uniquely in the form

$$x = b_{-k}p^{-k} + b_{-k+1}p^{-k+1} + b_{-k+2}p^{-k+2} + \cdots$$

for some $k > 0$, where

$$b_{-k} \neq 0, \quad b_i \in \{0, 1, \dots, p-1\}.$$

More generally, every element of $\mathbb{Q}_p$ admits a unique p -adic expansion with only finitely many negative powers of p .

HW:

Show that a polynomial $f \in \mathbb{Z}[x]$ has a root in $\mathbb{Z}_p$ if and only if for every $n \geq 1$, the congruence

$$f(x) \equiv 0 \pmod{p^n}$$

has a solution modulo p^n .